Regulatory Screening Parameters & Standard Customization Guide

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Challenge: Since AI Hackathon 2026 — Traficom: *"Saving Earth from Space Interferers Using Radio Frequencies"*
Applies to: generate_full_predictions.ps1 & gemini-code-1794028691323.html

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1. Overview: Standard Screening Frameworks

The screening pipeline evaluates all three regulatory frameworks simultaneously:

Standard / Framework	Limit Value	Regulatory Context & Application
Regulation M4	$-144.0\ \mathrm{dBW/m^2/4kHz}$	Traficom national baseline protection for **Fixed Links / Terrestrial Services** (corresponds to high-angle $> 25^\circ$ ITU limit).
Regulation M15	$-154.0\ \mathrm{dBW/m^2/4kHz}$	Traficom national protection for **Land-Mobile Services** (corresponds to strict low-angle $0^\circ \leq \delta \leq 5^\circ$ grazing threshold).
ITU RR Article 21	**Dynamic Table 21-4 Mask**	Dynamically interpolated by frequency (1.5 GHz to Ka-band) and topocentric elevation angle δ (0° to 90°).

In the generated Excel workbooks (`.xlsx`) and CSV files, **all three margins are calculated and reported side-by-side** in dedicated columns for every carrier-to-station link:

- `Worst_Margin_M4_dB (-144)`
- `Worst_Margin_M15_dB (-154)`
- `Worst_Margin_ITU_Art21_dB`

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2. Selecting the Standard via CLI in PowerShell

The PowerShell script (full_predictions/generate_full_predictions.ps1) supports a direct `-Standard` parameter:

```powershell
# 1. Default: AUTO (Dual check: M4 baseline + ITU Article 21 dynamic angle mask)
.\generate_full_predictions.ps1

# 2. Evaluate strictly against Regulation M15 (-154 dBW/m²/4kHz land-mobile limit)
.\generate_full_predictions.ps1 -Standard M15

# 3. Evaluate strictly against Regulation M4 (-144 dBW/m²/4kHz fixed service baseline)
.\generate_full_predictions.ps1 -Standard M4

# 4. Evaluate strictly against ITU Radio Regulations Article 21 Table 21-4
.\generate_full_predictions.ps1 -Standard ITU21
```

> **Note:** Regardless of which standard is selected to dictate the primary disposition flag (`CRITICAL_EXCEEDANCE` vs `COMPLIANT`), all three margin calculations remain intact in the Excel/CSV outputs.

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3. Key Lines to Tweak in the PowerShell Script (`generate_full_predictions.ps1`)

Lines	Purpose	What Can Be Tweaked
46–48	Parameter definition & default mode	`[ValidateSet("AUTO", "M4", "M15", "ITU21")]` and `$Standard = "AUTO"`.
559–611	`Get-Itu21Threshold` function	Defines band frequencies (1525 MHz, 2500 MHz, C-band, Ku-band, Ka-band) and elevation angle break-points (5° and 25°).
1290–1294	Margin calculation formulas	Change fixed dB values (`-144.0` for M4, `-154.0` for M15, or reference bandwidths).
1301–1314	Exceedance & Potential Concern trigger rules	The `$concernPrimary` condition (`margin >= -6.0 dB`). Adjust `-6.0` to change the proximity warning buffer (e.g., `-3.0 dB` or `-10.0 dB`).
1316–1325	Wi-Fi / Bluetooth downgrade policy	Handles license-exempt ISM/Wi-Fi bands (`LOW_SEVERITY_WIFI_BT`).

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4. Key Lines to Tweak in the JavaScript / UI Code (`gemini-code-1794028691323.html`)

A. Standard Selection Dropdowns in HTML:

- **Line 174–179**: `#rule-select` (Tab 1 Single Screening dropdown: `-144` [M4], `auto_itu21` [Table 21-4], `-154` [M15], `-126` [Art 21 hard limit]).
- **Line 496–501**: `#agent-standard-select` (Tab 2 Multi-Notice Screening dropdown).

- **Line 654–659**: `#cron-standard-select` (Tab 2 Automated Scheduler dropdown).
- **Line 748–753**: `#terr-standard-select` (Tab 4 Terrestrial Links dropdown: `M15`, `M4`, `Art21`).

B. Calculation Engine Class (`RegulatoryScreeningEngine`):

- **Line 1145–1170**: `getArticle21DynamicLimit(freqMHz, elevationDeg, refBwKHz)`
  - Baseline low-angle PFD values:
    - Below 1525 MHz: `lowLimit4kHz = -154.0` (flagged as `<1525 MHz` coordination trigger).
    - 1525–2500 MHz: `lowLimit4kHz = -154.0`
    - 2500–8500 MHz: `lowLimit4kHz = -152.0`
    - Ku-Band: `lowLimit4kHz = -150.0` / `-148.0`
    - Ka-Band (>15.4 GHz): `lowLimit4kHz = -139.0`
- **Line 1176–1185**: Elevation angle dynamic interpolation:
  - $\delta \leq 5^\circ$: $+0$ dB (strictest low-angle limit)
  - $5^\circ < \delta \leq 25^\circ$: $+0.5 \times (\delta - 5)$ dB
  - $\delta > 25^\circ$: $+10$ dB (plateau limit)
- **Line 1213–1247**: `evaluateRegulatoryMargin(calculatedPFD, thresholdPFD, isWifiBtBand)`:
  - `marginDB = calculatedPFD - thresholdPFD;`
  - `rawExceeded = marginDB > 0;` (Critical exceedance)
  - `rawConcern = marginDB >= -6.0;` (Potential concern proximity threshold)
  - Wi-Fi/Bluetooth downgrade gate when `#cron-downgrade-wifi` is active.